STAT 479: HW3

Spring 2026

Submission Details

Upload your solution to Gradescope as a PDF document. Include all relevant figures within your explanations. Include all relevant code. You may use either R or python.

Evaluation Criteria

Technical accuracy: Definitions used correctly, claims stated precisely.

Completeness: Assertions grounded in readings, lectures, or original code.

Clarity: Well-formatted, grammatically correct prose (not bullet fragments). No wordiness.

Code quality: Modular functions, meaningful variable names.

Problem 1 [Tree Ensemble Short Answer]

a. Imagine working on a large dataset at a company strong parallel computing infrastructure. Can the trees f^b in boosting be trained in parallel? What about bagging?

b. As you increase the number of trees B, do you expect the model to overfit, underfit, or "it depends"? Answer for both boosting and bagging.

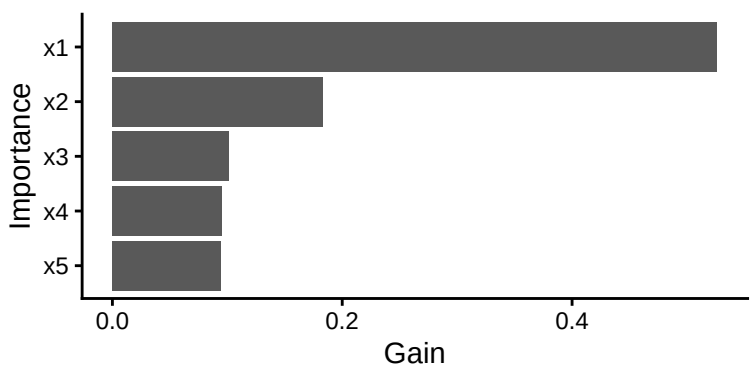
Problem 2 [Tree Ensemble Variable Importance]

You are working with a colleague on a classification problem using boosting. This is the data:

```r
df <- read_csv("https://github.com/krisrs1128/stat479_notes/raw/refs/heads/master/data/boosti
ix <- sample(nrow(df), 1600)
train <- df[ix, ]
test <- df[-ix, ]
```

You run an initial experiment on the dataset below and find these variable importances.

```r
task <- as_task_classif(train, target = "y")
learner <- lrn("classif.xgboost", nrounds = 100)
learner$train(task)
xgb.importance(model = learner$model) |>
    ggplot(aes(Gain, reorder(Feature, Gain))) +
    geom_col() +
    labs(x = "Gain", y = "Importance")
```



Your colleague runs the same experiment, but accidentally omitted `x1`, the variable with the highest importance. You compare the test classification error for both models.

```r
task_colleague <- as_task_classif(select(train, -x1), target = "y")
learner_colleague <- lrn("classif.xgboost", nrounds = 100)
learner_colleague$train(task_colleague)
tibble(
    model = c("full", "colleague"),
    classif.ce = c(
        learner$predict_newdata(test)$score(msr("classif.ce")),
        learner_colleague$predict_newdata(select(test, -x1))$score(msr("classif.ce"))
    )
)
```

```
# A tibble: 2 x 2
```

```
  model     classif.ce
  <chr>          <dbl>
1 full           0.162
2 colleague      0.158
```

Surprisingly, the performance has gone up! What do you think is going on, and what are the interpretability implications?

Problem 3 [LinearSHAP Derivation]

For certain models f and choices of v, SHAP values have closed-form expressions. Suppose f is linear and v uses baseline feature removal:

$$f(x) = \beta_0 + \sum_{j=1}^{J} \beta_j x_j, \qquad v(S) = f(x_S, x'_{\bar{S}})$$

a. Express $v(S)$ in terms of the β_j, the observed features x_j, and the baseline features x'_j.

b. Use your answer from (a) to find an expression for the marginal contribution $C(j \mid S)$.

c. Show that $\phi_j(f, x) = \beta_j(x_j - x'_j)$.

d. Interpret this result. How does the attribution depend on the baseline? Why might some observations x have larger attributions for feature j than others?

Problem 4 [Bikesharing Demand]

The bike demand dataset records hourly rentals along with weather, time-of-day, and seasonality features. The goal here is to predict hourly demand (`cnt`) and interpret the resulting models.

```r
f <- tempfile()
download.file("https://github.com/krisrs1128/treelapse/raw/master/data/bike.rda", f)
bike <- get(load(f)) |>
select(-dteday) |>
    mutate(
    across(where(is.character), as.factor)
)
```

a. Setup a boosting model pipeline that supports tuning over `nrounds`, `max_depth`, and `eta`. Use the ranges: `nrounds` $\in [50, 500]$, `max_depth` $\in [1, 6]$, and `eta` $\in [0.01, 0.3]$. Report the best configuration and its cross-validated RMSE. *(Note: Because these data were gathered over time, temporal cross-validation would be more accurate, but we're setting that aside since it's outside the scope of the class.)*

b. Retrain using the full data and the selected hyperparameters. Plot predicted vs. actual demand. Comment on any systematic patterns you notice.

c. Has the model captured a temperature effect (`temp`)? An interaction between `hr` and `weekend`? Answer both questions using CP profiles or SHAP attributions. Support your answer with 1 - 2 figures.

d. Fit an alternative model (e.g., lasso, CART, or bagging) to the same training data. You do not need to tune hyperparameters, but compute either CP profiles or SHAP attributions on your fit. Where do the models agree/disagree?

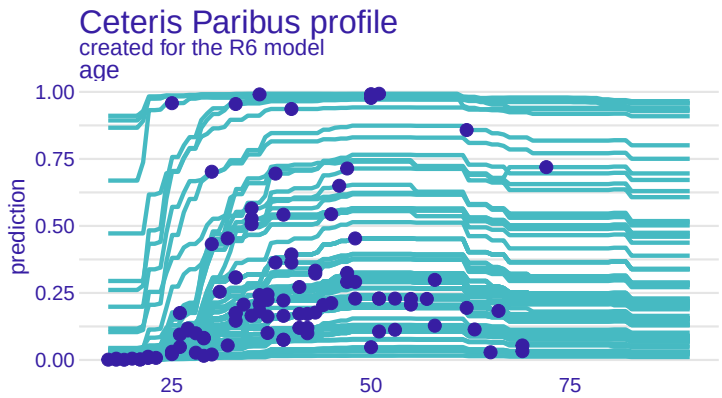
Problem 5 [CP vs. SHAP Visualizations]

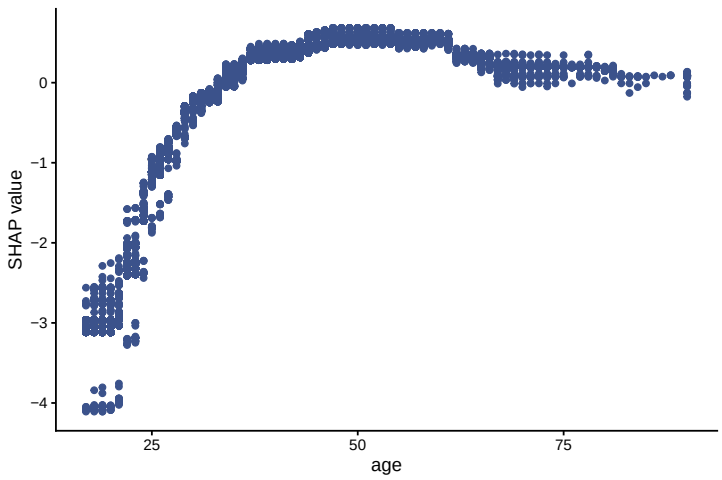
Both CP profiles and SHAP dependence plots show how a single feature relates to model predictions, but using different mathematical machinery. This problem compares the approaches using the census income dataset. As in our notes, we'll train a boosting classifier to predict whether a person earns more than $50K/year.

```
census <- read_csv("https://github.com/krisrs1128/stat479_notes/raw/refs/heads/master/data/c
task <- as_task_classif(census, target = "income")
task$positive <- ">50K"

learner <- lrn("classif.xgboost",
    nrounds = 200, eta = 0.05, max_depth = 3, predict_type = "prob"
)
learner$train(task)
```

The figures below show CP profiles and a SHAP dependence plot for `age`.





a. In the CP plot, what quantity does each curve trace out as `age` varies? In the SHAP plot, what does each point represent? Answer using notation from the lecture notes.

b. How does the efficiency axiom constrain the attributions you see in the SHAP dependence plot? Compared to the CP plot, does this change how you would explain the age-income effect to a non-statistical stakeholder?

c. Using either CP or SHAP, produce one additional interpretability plot for this model. It shoudn't just be the same code applied to a new feature. Include the figure and briefly discuss its strengths and weaknesses relative to the two plots above.